Ahmed Alaa

Unity Game Developer · gameplay systems and 3D art

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Gharbia, Egypt · Open to relocation (UAE, KSA)

SUMMARY

Unity Game Developer with 1+ year of professional experience and 2 years teaching Unity and C#. Shipped a Meta Quest 3 VR title and a bilingual (EN/AR) Photon PUN 2 multiplayer game, both internal releases for Zinad. Fits architecture to the project, from ScriptableObjects and state machines to direct references, favouring composition over inheritance. Models and textures his own art (Blender, Substance Painter). Hosted the game development session at NASA Space Apps Shebeen El-Kom. Experienced with Git, Agile/Scrum and AI-assisted development with Claude Code.

TECHNICAL SKILLS

Unity Engine — Unity 6, URP, Cinemachine, Addressables, Input System, Localization, Android & WebGL builds
Programming — C# (OOP), Lua, JavaScript (three.js), HTML & CSS
Architecture — Strategy, State & Object Pool patterns, Interface Contracts, ScriptableObject Architecture, Assembly Definitions, SOLID
Performance — Object pooling, allocation-free physics queries, event-driven systems over per-frame polling
Simulation & Math — Ballistics & Fire Control, Lead Targeting & Interception
Multiplayer & Data — Photon PUN 2, Firebase Realtime Database, JSON Data Management, RTL Localization (Arabic/English)
Tools — Git, GitHub, GitLab, Fork, Blender, Substance Painter, Premiere Pro, Photoshop, Roblox Studio (6 mo.)

PROFESSIONAL EXPERIENCE

Junior Unity Game Developer, Zinad 07/2025 – Present | Remote

- Shipped a VR rhythm-action game for Meta Quest 3 — migrated from Oculus SDK 0.46.0 to Meta XR All-in-One SDK 72.0.0, rebuilt the player prefab using Building Blocks, and resolved Android build failures caused by a TMP3D shader incompatibility with GLES3 multiview stereo rendering
- Implemented dual-saber hit detection with a same-frame double-hit guard via Script Execution Order, and a DSP clock-based spawner syncing object pool releases to BPM
- Built a ScriptableObject-based state machine handling gameplay/quiz transitions with fully decoupled cross-system communication, and a quiz system with DOTween crossfade transitions and live Arabic/English locale switching
- Shipped ZiRunner (internal release), a bilingual (EN/AR) multiplayer endless runner for security-awareness training — outcome-authoritative Photon PUN 2 netcode, RTL Arabic localization, and a StreamingAssets JSON content pipeline editable post-ship without a rebuild
- Built a real-time Firebase inventory system (learning project) as a reusable backend layer

Game Development Instructor, Da3em Academy 05/2023 – 08/2025 | Remote

- Designed and taught a full Unity and Roblox curriculum, mentoring students from first concept to their first launched games

PROJECTS

MilSim — Military Simulation Framework
Modular Unity 6 framework of military systems (C-RAM air defence, drone swarms, 2A65 Msta-B howitzer) that interact only through shared contracts, with assembly-enforced boundaries. Plugged a howitzer's shells into the air defence without editing it, then measured, diagnosed and fixed crossing-shot leaks: raised interception from 76% / 67% to 100%.

Emplacement · github.com/Ahmed-A-Abdou/emplacement
Open-source Unity package (MIT, UPM): StateMachine<T>, ObjectPool<T>, cross-prefab references. Extracted from MilSim; reused unchanged in a second project.

7 Knocks · abualaa.itch.io/7-knocks
Solo horror game for the Official 2026 Unity Game Jam, made in a week and playable in the browser. Modelled every asset in Blender, textured in Substance Painter; set the scare architecture and directed the code with an AI assistant.

SPEAKING & LEADERSHIP

Host, Game Development Session — NASA Space Apps Challenge, Shebeen El-Kom 10/2026

- Delivered “Build Systems, Not Spaghetti”, a 50-minute session for beginners on data-driven architecture, state machines and the strategy pattern, with live Unity demos

CIS Team MU — Team Leader & Game Development Instructor 09/2023 – 09/2024

- Led one of the university's largest student organizations (700+ members), running 10+ large events and workshops
- Trained 50+ members in Unity and C# through game jams (15+ prototypes); spoke at Beyond Earth, MagiCoders and VICTORIS 4.0

EDUCATION

Bachelor's Degree in Software Engineering 09/2021 – 07/2025
Faculty of Computer and Information Sciences, Mansoura University

AWARDS

1st Place, Advanced Level — Game Development Challenge, ITI TechAdventure (ITI, Cairo University) 01/2025

LANGUAGES

Arabic (Native) · English (Fluent – C2)